# ✹ EmployeeHub — Project Overview & Detailed Explanation

**EmployeeHub** is a full-stack **Employment Task Management & Performance Tracking System** developed with **Python, Flask, SQLite/SQLAlchemy, and modern web technologies**.
It centralizes workforce operations—replacing fragmented spreadsheets and manual tracking with a single, intelligent digital platform that provides real-time task monitoring, automated performance evaluations, attendance tracking, and data-driven workforce analytics.

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## 1. 🎯 Problem It Solves

| Traditional / Fragmented Approach | EmployeeHub Solution |
|---|---|
| Employee records scattered across spreadsheets | **Centralized CRUD database** with search, department filtering, and profiles |
| Tasks tracked via email/chat without progress tracking | **Interactive Kanban Board** with priority tags, overdue detection, and status workflows |
| Subjective, biased annual performance reviews | **Automated Weighted Performance Algorithm** based on tangible data |
| Manual attendance registers | **Digital daily attendance tracking** with monthly summaries |
| Tedious manual report compiling | **One-click PDF generation** (ReportLab) & **CSV export** (Pandas) |

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## 2. 🏛 System Architecture & Technology Stack

The application follows the **MVC (Model-View-Controller)** pattern with a dedicated **Service Layer** for clean separation of concerns:

```
EmployeeHub/
├── wsgi.py / app.py            # Production WSGI & Local entry points
├── config.py                   # App configurations (Database URI, Secret Keys)
├── requirements.txt            # Python dependencies (Flask, Gunicorn, Pandas, ReportLab)
├── app/
│   ├── models/                 # SQLAlchemy Database Models (7 tables)
│   ├── routes/                 # Flask Blueprints (Auth, Dashboard, Tasks, etc.)
│   ├── services/               # Core Business Logic (Performance, Insights, Reports)
│   ├── templates/              # Jinja2 Dynamic HTML Pages (18+ templates)
│   └── static/                 # CSS (Custom theming + Dark mode), JS, Icons
```

```
└── tests/                    # Automated Unit & Integration Test Suite (31 tests)
```

### Technology Breakdown

- **Backend**: Python 3, Flask 3.0 (Application Factory pattern with 8 modular Blueprints)
- **Database & ORM**: SQLite with SQLAlchemy ORM (7 relational tables)
- **Security & Authentication**: `Flask-Login` for session management and `Werkzeug` for salted password hashing
- **Frontend & UI**: HTML5, CSS3, JavaScript, Bootstrap 5, Bootstrap Icons, with a persistent **Dark Mode** switch
- **Data Visualization**: Chart.js (interactive doughnut, bar, and line charts)
- **Reporting & Data Export**: **ReportLab** for custom PDF report generation and **Pandas** for structured CSV data exports

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## 3. 🧩 Key Modules & Features

### 👤 1. Employee Management Module

- Complete CRUD operations (Add, Edit, View, Deactivate employees).
- Detailed employee profile cards displaying personal info, department, designation, assigned manager, performance history, and active tasks.
- Instant search and multi-parameter filtering (by department, status, role).

### 📋 2. Task Management & Kanban Board

- Visual Kanban columns: **To Do**, **In Progress**, **Under Review**, and **Completed**.
- Priority levels: *Low, Medium, High, Urgent*.
- Automatic overdue detection if due date has passed.
- Quick status drag-and-drop or modal updates.

### 📈 3. Performance Tracking & Weighted Scoring Algorithm

Rather than subjective reviews, performance is calculated using a **transparent mathematical formula**:

$$\text{Final Score} = (\text{Task Completion} \times 0.30) + (\text{Quality Score} \times 0.25) + (\text{Attendance} \times 0.15) + (\text{Productivity} \times 0.20) + (\text{Manager Rating} \times 0.10)$$

Performance Classifications:

| Score Range | Classification Tier |
| --- | --- |
| **90 – 100** | 🌟 **Excellent** |
| **80 – 89** | 🟢 **Very Good** |
| **70 – 79** | 🔵 **Good** |

| Score Range | Classification Tier |
|---|---|
| **60 – 69** | 🟡 **Average** |
| **Below 60** | 🔴 **Needs Improvement** |

## 📅 4. Attendance Monitoring

- Daily check-in / check-out status: *Present, Absent, On Leave, Late*.
- Monthly percentage calculations and department-wise attendance averages.

## 🏆 5. Gamification & Leaderboard

- Dynamic ranking of employees based on monthly scores.
- Automated achievement badges: *Top Performer*, *Productivity Star*, *Quality Champion*.

## 🧠 6. Automated Rule-Based Insights

- Automatically analyzes data trends to generate proactive management alerts:
  - Identifies top achievers
  - Flags productivity drops compared to previous periods
  - Detects employees with high risk of missing deadlines

## 📄 7. Reports & Data Export

- **PDF Report Generation**: Uses ReportLab to generate formatted employee dossiers and performance summaries.
- **CSV Data Export**: Uses Pandas for data extraction into downloadable spreadsheets.

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# 4. 👥 Role-Based Access Control (RBAC)

| Feature / Action | Admin | Employee |
|---|---|---|
| View Organization Dashboard & Analytics | ✅ | ❌ (Own dashboard only) |
| Manage / Add / Edit Employees | ✅ | ❌ |
| Create & Assign Tasks | ✅ | ❌ |
| View & Update Assigned Tasks | ✅ | ✅ |
| Mark Daily Attendance | ✅ | ✅ (Own attendance) |
| Download PDF Reports & CSV Exports | ✅ | ✅ (Own performance report) |

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# 5. 🗄 Database Design (7 Relational Tables)

1. `users`: Authentication credentials, hashed passwords, and assigned roles.
2. `employees`: Personal info, job designation, joining date, avatar colors, status.
3. `departments`: IT, HR, Finance, Marketing, Operations, department heads.
4. `tasks`: Title, description, priority, progress %, due dates, quality scores.
5. `performances`: Historical monthly scores, metrics breakdown, classification.
6. `attendances`: Daily records, check-in timestamps, status.
7. `feedbacks`: Peer-to-peer and manager feedback logs.

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# 6. 🧪 Quality & Testing

- Includes a complete **pytest test suite** (`tests/test_app.py`) with **31 automated tests** covering authentication, CRUD security, algorithm correctness, boundary conditions, and PDF exports.